

Formula Sheet

Constants

$$\vec{g} = 9.81 \frac{m}{s^2} [down]$$

$$G = 6.67 \times 10^{-11} \frac{N \cdot m^2}{kg^2}$$

$$c = 3.00 \times 10^8 \frac{m}{s}$$

$$e = 1.60 \times 10^{-19} C$$

$$m_e = 9.109 \times 10^{-31} kg$$

$$m_p = 1.673 \times 10^{-27} kg$$

$$h = 6.63 \times 10^{-34} J \cdot s$$

$$k = 8.99 \times 10^9 \frac{N \cdot m^2}{C^2}$$

$$\mu_{air} = 4\pi \times 10^{-7} \frac{Tm}{A}$$

Miscellaneous

$$slope = \frac{rise}{run} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$quadratic \text{ formula for } ax^2 + bx + c = 0 \rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

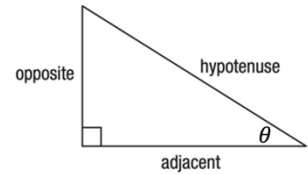
$$Pythagorean \text{ theorem (right angle triangle)} \rightarrow c^2 = a^2 + b^2$$

$$trigonometry \text{ (right angle triangle)} \rightarrow \sin \theta = \frac{opp}{hyp}; \cos \theta = \frac{adj}{hyp}; \tan \theta = \frac{opp}{adj}$$

$$sine \text{ law} \rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$cosine \text{ law} \rightarrow c^2 = a^2 + b^2 - 2ab \cos C$$

$$\%error = \left| \frac{theoretical \text{ value} - experimental \text{ value}}{theoretical \text{ value}} \right| \times 100\%$$



Dynamics

$$\vec{v} = \frac{d}{t}$$

$$\vec{v}_2 = \vec{v}_1 + \vec{a}t$$

$$\Delta \vec{d} = \left(\frac{\vec{v}_1 + \vec{v}_2}{2} \right) t$$

$$\Delta \vec{d} = \vec{v}_1 t + \frac{1}{2} \vec{a} t^2$$

$$\Delta \vec{d} = \vec{v}_2 t - \frac{1}{2} \vec{a} t^2$$

$$(\vec{v}_2)^2 = (\vec{v}_1)^2 + 2\vec{a}\Delta \vec{d}$$

$$\vec{F}_{net} = m\vec{a}$$

$$\vec{F}_{net} = \Sigma \vec{F}$$

$$\vec{F}_g = m\vec{g}$$

$$\vec{F}_k = \mu_k \vec{F}_N$$

$$\vec{F}_s \leq \mu_s \vec{F}_N$$

$$F_g = \frac{Gm_1m_2}{d^2}$$

Circular Motion

$$\tan \theta = \frac{v^2}{rg}$$

$$f = \frac{1}{T}$$

$$a_c = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2} = 4\pi^2 r f^2$$

$$F_c = \frac{mv^2}{r} = \frac{4\pi^2 rm}{T^2} = 4\pi^2 r m f^2$$

Energy & Momentum

$$W = F\Delta d \cos \theta$$

$$E_g = mg\Delta h$$

$$E_k = \frac{1}{2} mv^2$$

$$\%efficiency = \frac{E_{out}}{E_{in}} \times 100\%$$

$$P = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$W = \Delta E$$

$$E_e = \frac{1}{2} kx^2$$

$$\vec{F}_x = -k\Delta \vec{x} \quad T = 2\pi \sqrt{\frac{m}{k}}$$

$$E_k + E_g + E_e = E'_k + E'_g + E'_e$$

$$\vec{p} = m\vec{v}$$

$$\overrightarrow{\Delta p} = \vec{F}\Delta t = m\vec{v}_2 - m\vec{v}_1$$

$$elastic \text{ collision} \rightarrow m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}'_1 + m_2 \vec{v}'_2; \quad \text{if } \vec{v}_2 = 0 \rightarrow \vec{v}'_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_1; \quad \vec{v}'_2 = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_1$$

$$inelastic \text{ collision} \rightarrow m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}'$$

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Gravitational, Electric & Magnetic FieldsGravitational Fields

$$F_g = \frac{Gm_1m_2}{r^2} \quad E_g = \frac{-Gm_1m_2}{r} \quad g = \frac{Gm}{r^2} \quad v_{\text{escape}} = \sqrt{\frac{2Gm}{r}} \quad v_{\text{orbital}} = \sqrt{\frac{Gm}{r}} = \frac{2\pi r}{T}$$

Electric Fields

$$\text{near a point charge} \rightarrow \epsilon = \frac{F_E}{q} = \frac{kq}{r^2}; V = \frac{kq}{r} \quad \text{point charges} \rightarrow F_E = \frac{kq_1q_2}{r^2}; E_E = \frac{kq_1q_2}{r}$$

$$\text{electric field} \rightarrow \Delta E_E = -q\epsilon\Delta d; V = \frac{E_E}{q}; \epsilon = -\frac{\Delta V}{\Delta d} \quad \text{Millikan oil drop} \rightarrow q = \frac{mg\Delta d}{\Delta V}$$

Magnetic Fields

$$\text{on charged particle} \rightarrow F_M = qvB\sin\theta \quad \text{on straight conductor} \rightarrow F_M = IlB\sin\theta$$

$$I = \frac{q}{\Delta t} \quad B = \frac{\mu I}{2\pi r} \quad B = \frac{\mu NI}{L}$$

The Wave Nature of Light

$$v = f\lambda \quad f = \frac{1}{T} = \frac{N}{\Delta t} \quad n_1\sin\theta_1 = n_2\sin\theta_2 \quad \frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \quad n = \frac{c}{v}$$

$$\text{double-slit interference (constructive)} \rightarrow m\lambda = d\sin\theta; \quad x_m = \frac{mL\lambda}{d}$$

$$\text{double-slit interference (destructive)} \rightarrow \left(n - \frac{1}{2}\right)\lambda = d\sin\theta; \quad x_n = \left(n - \frac{1}{2}\right)\frac{L\lambda}{d}$$

$$\text{thin film interference} \rightarrow \text{Path Difference} = \frac{2tn_2}{\lambda}$$

$$\text{single-slit interference (constructive)} \rightarrow \left(m + \frac{1}{2}\right)\lambda = w\sin\theta_m$$

$$\text{single-slit interference (destructive)} \rightarrow n\lambda = w\sin\theta_n; \quad y = \frac{\lambda L}{w}; \quad \text{central max} \rightarrow 2\Delta y$$

$$\text{Brewster's Law} \rightarrow \tan\theta_B = \frac{n_2}{n_1} \quad \text{Malus' Law} \rightarrow I_{\text{out}} = I_{\text{in}}\cos^2\theta$$

Modern Physics

$$\Delta t_m = \frac{\Delta t_s}{\sqrt{1-\frac{v^2}{c^2}}} \quad L_m = L_s\sqrt{1-\frac{v^2}{c^2}} \quad m_{\text{relativistic}} = \frac{m_{\text{rest}}}{\sqrt{1-\frac{v^2}{c^2}}} \quad p = \frac{mv}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$E_{\text{rest}} = mc^2 \quad E_{\text{total}} = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} \quad E_{\text{total}} = E_{\text{rest}} + E_k$$

$$E_{\text{photon}} = hf = \frac{hc}{\lambda} \quad W = e\Delta V \quad E_{k(\text{max})} = E_{\text{photon}} - W$$

$$p = \frac{hf}{c} \quad \lambda = \frac{h}{p} = \frac{h}{mv}$$